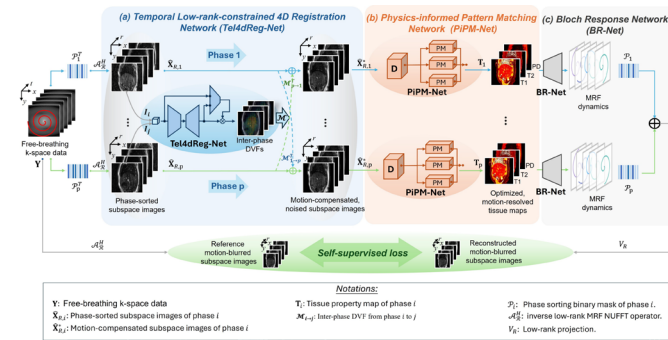


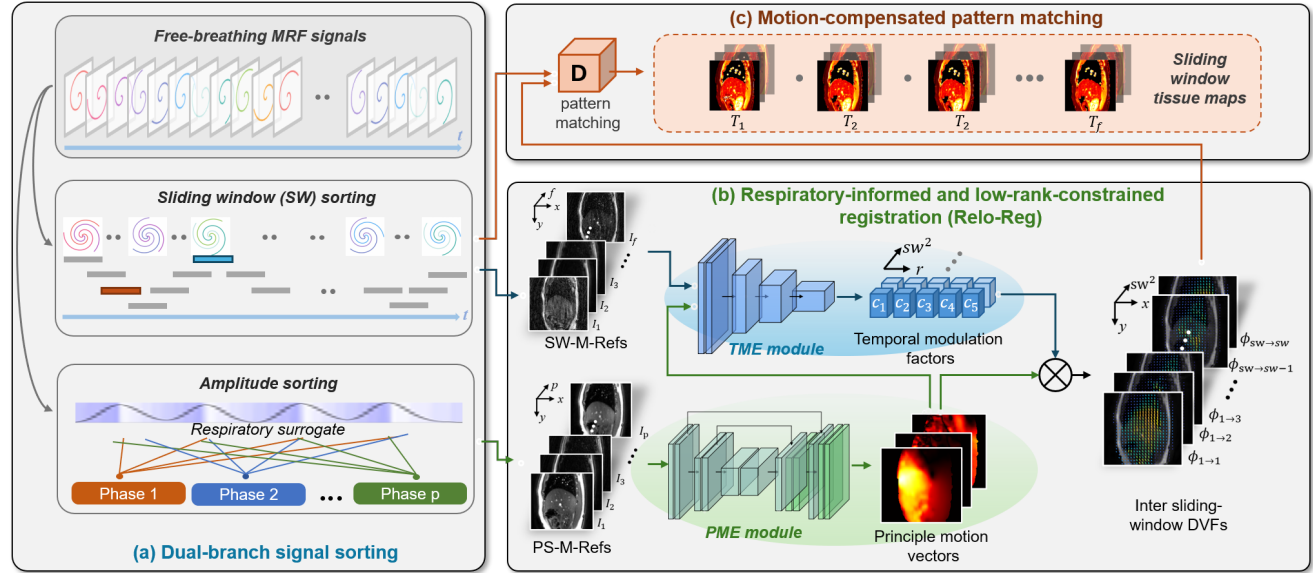
1. Background/Motivation

- Current time-resolved MRF methods for liver typically employ a **retrospective sorting approach**. However, this strategy is limited in patients with irregular breathing patterns, as it cannot fully capture the true respiratory dynamics and may introduce binning artifacts.
- There is a growing need for time-resolved cine MRF (cMRF-T) technique that characterizes respiratory motion in its natural temporal order.



Conventional retrospective sorted time-resolved reconstruction.

2. Time-resolved Cine MR Fingerprinting (cMRF-T) framework



Schematic overview of the time-resolved cine MRF (cMRF-T) technique, which incorporates three key components, including dual-branch signal sorting to process free-breathing MRF signals, respiratory-informed and low-rank constrained registration (Relo-Reg) to calculate motion and motion-compensated pattern matching to obtain motion-resolved tissue maps.

3. Relo-Reg Algorithm

- Principal motion estimation (PME) module**
Captures the dominant deformation patterns from the phase-specific motion reference images (PS-M-refs).
- Temporal modulation estimation (TME) module**
Adapts dominant patterns to generate inter-sliding window DVFs guided by the sliding window motion reference images.
- Self-supervised loss**
Local correlation coefficient (LCC) loss will compute all the registration misalignment between every two sliding window motion reference images.

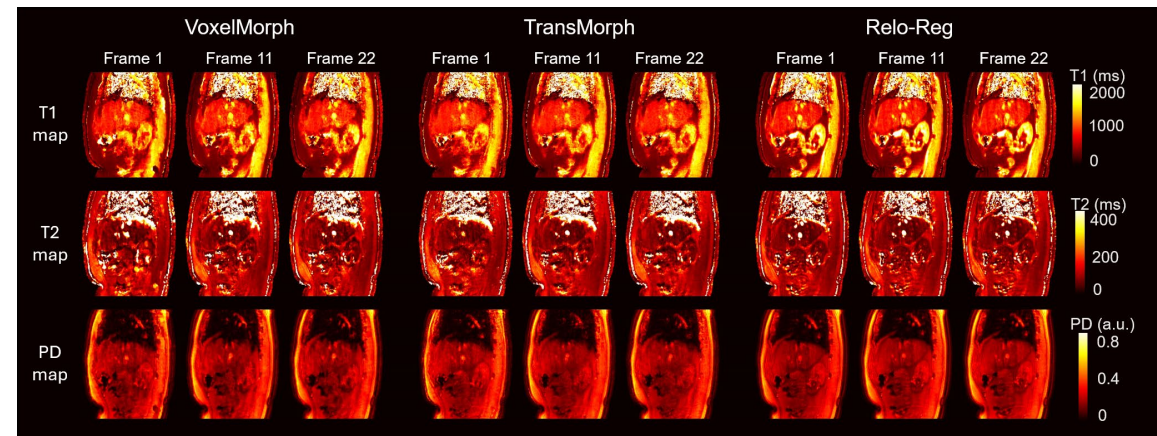
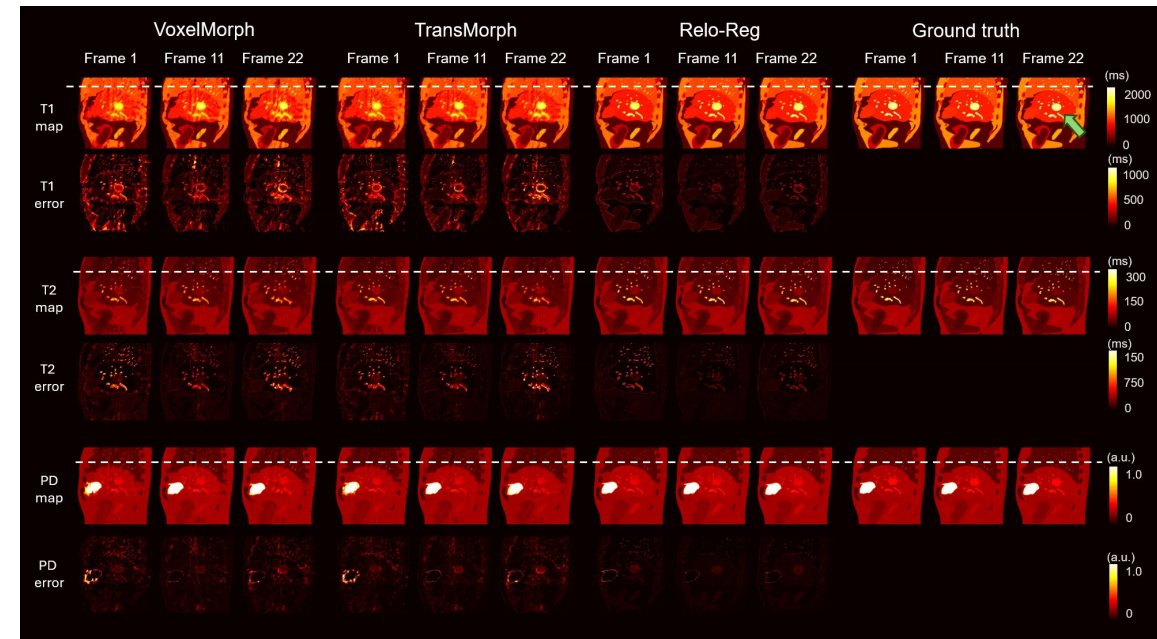
5. Quantitative Results

		T1	T2	PD
Liver	VoxelMorph	0.2042 ± 0.0571	0.6284 ± 0.1965	0.1492 ± 0.1048
	TransMorph	0.1750 ± 0.0641	0.4621 ± 0.2047	0.1427 ± 0.1128
	Relo-Reg	0.0909 ± 0.0346	0.2859 ± 0.1041	0.106 ± 0.0995
Tumor	VoxelMorph	0.2522 ± 0.0732	0.2819 ± 0.1024	0.1867 ± 0.1046
	TransMorph	0.2569 ± 0.0826	0.2412 ± 0.0822	0.1837 ± 0.1104
	Relo-Reg	0.1120 ± 0.0564	0.1385 ± 0.0588	0.0981 ± 0.0963
Kidney	VoxelMorph	0.1490 ± 0.0379	0.1707 ± 0.0561	0.1116 ± 0.0243
	TransMorph	0.1438 ± 0.047	0.1435 ± 0.0553	0.1089 ± 0.0269
	Relo-Reg	0.0722 ± 0.0219	0.0757 ± 0.0267	0.0635 ± 0.0176
Muscle	VoxelMorph	0.0970 ± 0.0199	0.1474 ± 0.0581	0.1161 ± 0.1013
	TransMorph	0.0958 ± 0.0193	0.1506 ± 0.0624	0.1264 ± 0.1115
	Relo-Reg	0.0599 ± 0.0195	0.0888 ± 0.0398	0.0786 ± 0.0979
Gall bladder	VoxelMorph	0.0789 ± 0.0334	0.1112 ± 0.0431	0.0879 ± 0.0233
	TransMorph	0.0885 ± 0.0336	0.0983 ± 0.0437	0.1058 ± 0.0311
	Relo-Reg	0.0616 ± 0.0213	0.1107 ± 0.0464	0.0566 ± 0.026
Fat	VoxelMorph	0.4274 ± 0.2243	0.1564 ± 0.1144	0.1557 ± 0.1153
	TransMorph	0.4317 ± 0.2621	0.1692 ± 0.1367	0.1566 ± 0.1235
	Relo-Reg	0.2517 ± 0.1004	0.1011 ± 0.0528	0.0921 ± 0.0917

6. Acknowledgement

This research was supported by the National Natural Science Foundation of China (NSFC) Young Scientist Fund (82202941) and by Hong Kong funding schemes, including the General Research Fund (GRF 15104323, GRF 15102219, GRF 15104822), the Health and Medical Research Fund (HMRF 10211606, 12231546), the Early Career Scheme (ECS 25100525), and the Innovation and Technology Support Programme (ITS/049/22FP).

4. Qualitative Results



5. Conclusion /Take-Home Message

- In this study, we successfully developed a cMRF-T framework. This framework overcomes fundamental barriers in motion-resolved MRF that relied on retrospective sorting.
- cMRF-T enables robust visualization of respiratory motion under irregular motion pattern.
- Relo-Reg, while initially developed for 4DMRF reconstruction, holds broader promise for general 4D imaging applications in scenarios that require high-fidelity motion modeling from significantly under-sampled motion reference images.